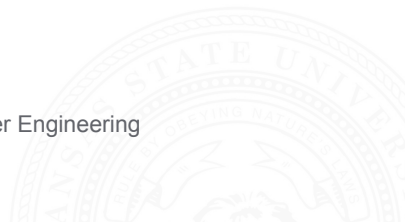


Per Unit System III

Single-phase transformers and three-phase Y–delta normalization

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Lecture outline

Example 1: single-phase circuit with transformers

- 1 Two transformer ratios, three voltage zones, and one complete PU solution.

Example 2: three-phase Y- Δ normalization

- 2 Physical Δ load $\rightarrow$ equivalent Y $\rightarrow$ PU model $\rightarrow$ branch quantities.

Each part follows the circuit and questions with two workflow pages, then the complete solution.

Part 1: single-phase voltage levels

Example 1

Source, two transformers, line, and load

Three voltage zones in one single-phase circuit

Transformer ratios connect the voltage bases. Per unit gives one series circuit.

Example 1: circuit and data

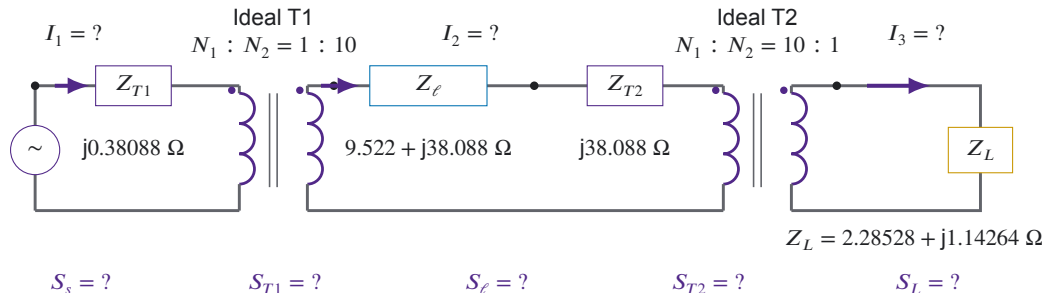
Single-phase RMS circuit: $S_{\text{base}} = 100 \text{ MVA}$, $V_{\text{base},1} = 13.8 \text{ kV}$. All impedances shown are physical ohms.

B1: $V_s = V_{B1}$
 $13.8 \angle 0^\circ \text{ kV}$

B2
 $V_{B2} = ?$

B3
 $V_{B3} = ?$

B4
 $V_{B4} = ?$



The leakage impedances are on the drawn primary sides. Dots define the voltage polarity; magnetizing branches are neglected.

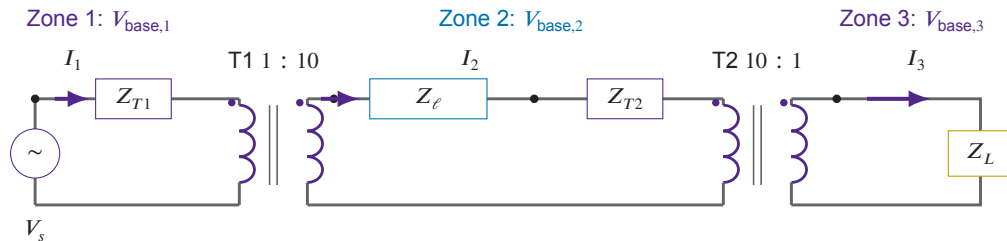
V_{Bk} is measured from the upper terminal to its local return. S_s is supplied; other powers are absorbed.

Example 1: questions

Goal: Solve the single-phase circuit and assess its load voltage.

- 1 Normalization.** Find compatible V_{base} , Z_{base} , I_{base} in all zones. Convert the source and all impedances to pu.
- 2 Current.** Draw the common-base circuit and calculate Z_{total} and $I_{\text{p.u.}}$ in rectangular and polar forms.
- 3 Voltages and amperes.** Find V_{B2} , V_{B3} , V_{B4} in pu and kV, then I_1 , I_2 , I_3 in amperes.
- 4 Power and decision.** Find S_s , S_{T1} , S_ℓ , S_{T2} , S_L in MVA and check their balance. Does $|V_{B4}| \geq 0.95$ p.u.?

Example 1 workflow: bases and current



1 Normalize each voltage zone.

Follow T1 and T2 to set voltage bases.
Use the local base for each impedance.

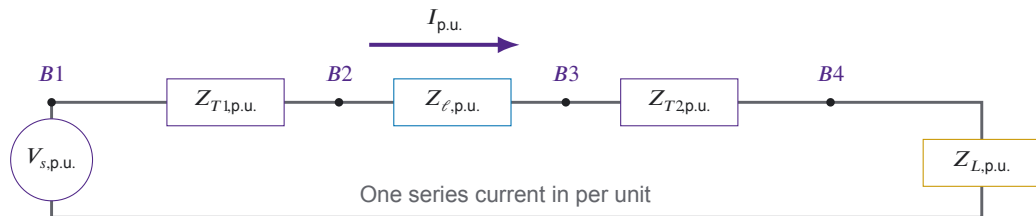
2 Find one per-unit current.

Compatible bases remove ideal ratios.
Sum the four series impedances.

$$Z_{base,k} = \frac{V_{base,k}^2}{S_{base}}, \quad Z_{k,p.u.} = \frac{Z_k}{Z_{base,k}}$$

$$I_{p.u.} = \frac{V_{s,p.u.}}{Z_{total,p.u.}}$$

Example 1 workflow: voltages and power



3 Follow B1 through B4.

Subtract each $I_{p.u.} Z_{p.u.}$ drop. Restore volts and amperes with the local bases.

$$V_B = V_{B,p.u.} V_{\text{base}}, \quad I_k = I_{p.u.} I_{\text{base},k}$$

4 Check power and load voltage.

Use $S_{s,p.u.} = V_{s,p.u.} I_{p.u.}^*$ and $S_{k,p.u.} = |I_{p.u.}|^2 Z_{k,p.u.}$.

$$S_s \stackrel{?}{=} S_{T1} + S_{\ell} + S_{T2} + S_L$$

Compare $|V_{B4,p.u.}|$ with 0.95.

E1 Step 1: single-phase bases

$$S_{\text{base}} = 100 \text{ MVA}, \quad Z_{\text{base},k} = \frac{V_{\text{base},k}^2}{S_{\text{base}}}, \quad I_{\text{base},k} = \frac{S_{\text{base}}}{V_{\text{base},k}}$$

Zone	V_{base} (kV)	Z_{base} (Ω)	I_{base} (A)
1: source / T1 primary	13.8	1.9044	7246.377
2: line / T2 primary	138	190.44	724.638
3: T2 secondary / load	13.8	1.9044	7246.377

The voltage bases follow the 1:10 and 10:1 turns ratios. This single-phase current base uses no $\sqrt{3}$ factor.

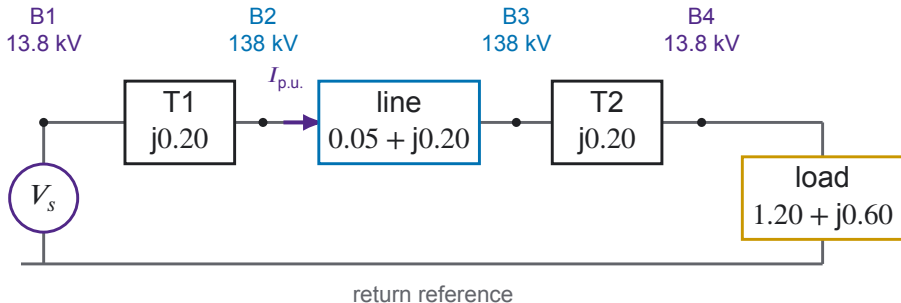
E1 Step 1: normalized data

$$V_{s,\text{p.u.}} = \frac{13.8\angle 0^\circ}{13.8} = 1\angle 0^\circ, \quad Z_{k,\text{p.u.}} = \frac{Z_k}{Z_{\text{base},k}}$$

Element	Physical impedance (Ω)	Base (Ω)	PU impedance
T1 leakage	j0.38088	1.9044	j0.20
Line	9.522 + j38.088	190.44	0.05 + j0.20
T2 leakage	j38.088	190.44	j0.20
Load	2.28528 + j1.14264	1.9044	1.20 + j0.60

The two leakage impedances differ in physical ohms but have the same per-unit value on their local bases.

E1 Step 2: PU circuit



Step 1 result: $Z_L = 1.20 + j0.60$ p.u.. **Step 2 uses this circuit to find $I_{p.u.}$.**

E1 Step 2: network impedance

$$Z_{\text{net}} = Z_{T1} + Z_{\ell} + Z_{T2},$$

$$Z_{\text{net}} = j0.20 + (0.05 + j0.20) + j0.20 = \boxed{0.05 + j0.60 \text{ p.u.}}$$

R_{net}

0.05 p.u.

The line contributes the complete series resistance.

X_{net}

0.60 p.u.

T1, line, and T2 each contribute 0.20.

Keep Z_{net} separate. It supports both voltage-drop and power checks.

E1 Step 2: total impedance

$$Z_{\text{total}} = Z_{\text{net}} + Z_L,$$

$$Z_{\text{total}} = (0.05 + j0.60) + (1.20 + j0.60) = \boxed{1.25 + j1.20 \text{ p.u.}}$$

Real part

0.05 network plus 1.20 load.

1.25 p.u.

Imaginary part

0.60 network plus 0.60 load.

1.20 p.u.

A positive total impedance angle predicts a lagging current.

E1 Step 2: series current

With $V_s = 1\angle 0^\circ$ and $Z_{\text{total}} = 1.25 + j1.20$ in per unit,

$$|Z_{\text{total}}| = \sqrt{1.25^2 + 1.20^2} = 1.732772, \quad \angle Z_{\text{total}} = 43.830861^\circ.$$

$$I_{\text{pu}} = \frac{1\angle 0^\circ}{1.732772\angle 43.830861^\circ} = 0.577110\angle(-43.830861^\circ).$$

For the voltage-drop calculations, use the same current in rectangular form:

$$I_{\text{pu}} = 0.416320 - j0.399667.$$

Step 2 result: $I_{\text{p.u.}} = 0.416320 - j0.399667$. **Use this value for every bus-voltage calculation.**

E1 Step 3: T1 drop

$$\Delta V_{T1} = I_{\text{p.u.}} Z_{T1} = (0.416320 - j0.399667)(j0.20),$$

$$\Delta V_{T1} = 0.079933 + j0.083264 \text{ p.u.}$$

Real-part check

$$-\text{Im}\{I\}X = 0.079933$$

Imaginary-part check

$$\text{Re}\{I\}X = 0.083264$$

Multiplication by positive jX rotates the lagging current by $+90^\circ$.

E1 Step 3: bus B2

$$V_{B2} = V_{B1} - \Delta V_{T1},$$

$$V_{B2} = 1 - (0.079933 + j0.083264) = \boxed{0.920067 - j0.083264 \text{ p.u.}},$$

$$\boxed{V_{B2} = 0.923827 \angle (-5.171052^\circ) \text{ p.u.}}$$

Bus B2 is in the 138-kV zone even though its per-unit voltage is computed in the same circuit.

E1 Step 3: line drop

$$I_{\text{p.u.}}(0.05) = 0.020816 - j0.019983, \quad I_{\text{p.u.}}(j0.20) = 0.079933 + j0.083264,$$

$$\Delta V_{\ell} = I_{\text{p.u.}}(0.05 + j0.20) = \boxed{0.100749 + j0.063281 \text{ p.u.}}$$

The R term follows the current angle.

The jX term is rotated by $+90^\circ$.

E1 Step 3: bus B3

$$V_{B3} = V_{B2} - \Delta V_{\ell},$$

$$V_{B3} = (0.920067 - j0.083264) - (0.100749 + j0.063281),$$

$$V_{B3} = 0.819317 - j0.146545 = 0.832320 \angle (-10.140793^\circ) \text{ p.u.}$$

The line reduces both the magnitude and angle of the downstream bus voltage.

E1 Step 3: bus B4

$$\Delta V_{T2} = I(j0.20) = 0.079933 + j0.083264, \quad V_{B4} = V_{B3} - \Delta V_{T2},$$

$$V_{B4} = (0.819317 - j0.146545) - (0.079933 + j0.083264),$$

$$V_{B4} = 0.739384 - j0.229808 \text{ p.u.}$$

Bus B4 is the load bus in Zone 3. Keep that zone label for physical back-conversion.

E1 Step 3: voltage check

Load-impedance route

$$V_L = I_{\text{p.u.}} Z_L$$
$$= \boxed{0.739384 - j0.229808 \text{ p.u.}}$$

Network-drop route

$$V_L = V_s - I_{\text{p.u.}} Z_{\text{net}}$$
$$= \boxed{0.739384 - j0.229808 \text{ p.u.}}$$

Agreement checks the current, the network reduction, and all three series drops at once.

E1 Step 3: load voltage

$$V_{B4,\text{p.u.}} = 0.739384 - j0.229808 = 0.774274 \angle (-17.265809^\circ)$$

$$V_{B4} = V_{B4,\text{p.u.}} V_{\text{base},3} = \boxed{10.684983 \angle (-17.265809^\circ) \text{ kV}}$$

This is the voltage across the single-phase load terminals. Multiply by the local voltage base directly.

E1 Step 3: actual currents

$$|I_{\text{p.u.}}| = 0.577110, \quad I_k = I_{\text{p.u.}} I_{\text{base},k}$$

Zone 2: high-voltage loop

$$I_{\text{base},2} = 724.637681 \text{ A}$$

$$I_2 = 418.196 \angle (-43.831^\circ) \text{ A}$$

Zones 1 and 3: low-voltage loops

$$I_{\text{base},1,3} = 7.246376812 \text{ kA}$$

$$I_1 = I_3 = 4.181955 \angle (-43.831^\circ) \text{ kA}$$

The physical currents change by the inverse turns ratio: $|I_1| = 10|I_2| = |I_3|$.

E1 Step 3: voltage results

Bus	Base (kV)	$ V $ (pu)	$\angle V$	$ V $ (kV)
B1	13.8	1.0000	0°	13.800
B2	138	0.9238	-5.171°	127.488
B3	138	0.8323	-10.141°	114.860
B4	13.8	0.7743	-17.266°	10.685

$$V_{Bk} = V_{Bk,\text{p.u.}} V_{\text{base,local}}$$

Every voltage is measured from the upper conductor to the local return shown in the physical circuit.

E1 Step 4: source power

$$S_s = V_s I_{p.u.}^* = (1\angle 0^\circ)(0.416319734 + j0.399666944),$$

$$S_s = 0.416319734 + j0.399666944 \text{ p.u.},$$

$$S_s = 41.631973 + j39.966694 \text{ MVA supplied}$$

Positive P_s and Q_s are supplied by the ideal source under the stated source convention.

E1 Step 4: absorbed powers

Single-phase absorbed powers on the 100-MVA base

$$|I_{p.u.}|^2 = 0.333055787, \quad S_k = |I_{p.u.}|^2 Z_k$$

Element	P_k (p.u.)	Q_k (p.u.)	P_k (MW)	Q_k (Mvar)
T1	0	0.0666	0	6.661
Line	0.0167	0.0666	1.665	6.661
T2	0	0.0666	0	6.661
Load	0.3997	0.1998	39.967	19.983

Line resistance absorbs real network power. Positive reactances absorb reactive power.

E1 Step 4: power balance

$$\sum S_{\text{absorbed}} = S_{T1} + S_{\ell} + S_{T2} + S_L,$$
$$\sum S_{\text{absorbed}} \approx 0.4163197336 + j0.3996669442 \text{ p.u.},$$

$$S_s - \sum S_{\text{absorbed}} = \epsilon_S, \quad |\epsilon_S| < 3 \times 10^{-16} \text{ p.u.}$$

Exact-input algebra closes at zero. The machine residual is numerical roundoff.

Example 1: final answer

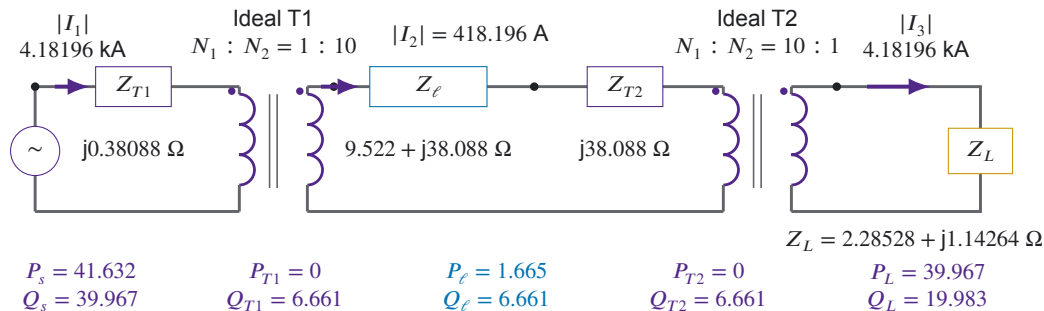
Same single-phase circuit; rounded values. Bus labels: pu / kV magnitude. P in MW; Q in Mvar.

$$\begin{aligned} \text{B1: } & 1 \angle 0^\circ \text{ p.u.} \\ |V_{B1}| &= 13.800 \text{ kV} \end{aligned}$$

$$\begin{aligned} \text{B2: } & 0.9238 \angle -5.2^\circ \text{ p.u.} \\ |V_{B2}| &= 127.488 \text{ kV} \end{aligned}$$

$$\begin{aligned} \text{B3: } & 0.8323 \angle -10.1^\circ \text{ p.u.} \\ |V_{B3}| &= 114.860 \text{ kV} \end{aligned}$$

$$\begin{aligned} \text{B4: } & 0.7743 \angle -17.3^\circ \text{ p.u.} \\ |V_{B4}| &= 10.685 \text{ kV} \end{aligned}$$



$I_{p.u.} = 0.577110 \angle (-43.831^\circ)$; all loop-current angles are -43.831° .

B4: $0.774274 < 0.95 \text{ p.u.}$. Source power equals the sum of absorbed powers (before rounding).

Part 2: three-phase Y- Δ normalization

Example 2

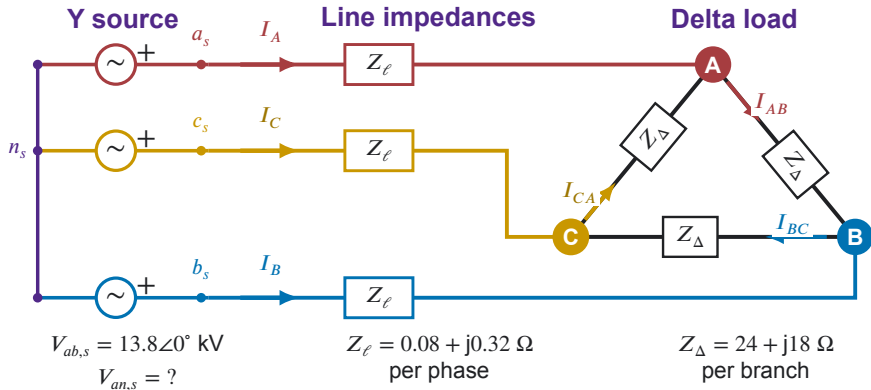
Y source, three line impedances, and a Δ load

An equivalent Y model for per-unit analysis

The equivalent circuit gives line quantities. The physical Δ gives branch quantities.

Example 2: circuit and data

Balanced abc RMS: $S_{\text{base},3\phi} = 20 \text{ MVA}$, $V_{LL,\text{base}} = 13.8 \text{ kV}$. Ideal Y source; constant-Z delta load.



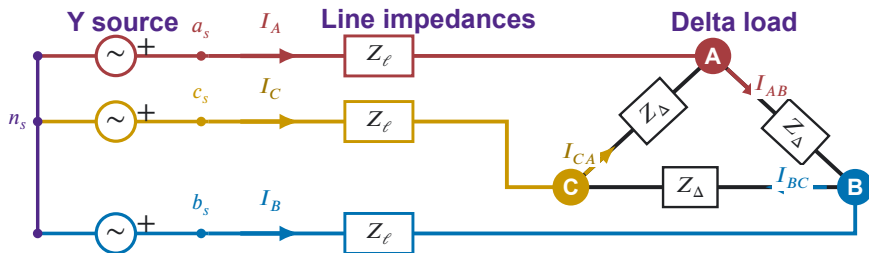
Transform the physical Δ to an equivalent Y before forming the per-phase PU circuit.

Example 2: questions

Goal: Normalize the given three-phase circuit and recover its physical Δ quantities.

- 1 Connection conversion.** Find the equivalent Z_Y for $Z_{\Delta} = 24 + j18 \Omega$. Show which terminal quantities are preserved.
- 2 Normalization.** Find $V_{LN,base}$, I_{base} , Z_{base} and normalize the phase source, line, and equivalent Y load.
- 3 PU solution.** Find I_A , the equivalent load voltage V_{AN} , and the physical load line voltage V_{AB} .
- 4 Physical Δ .** Find the branch currents and total load power. Verify $I_A = I_{AB} - I_{CA}$ and compare Y / Δ power results.

Example 2 workflow: connection and bases



- 1 Replace the load by an equivalent Y.

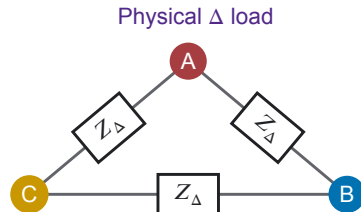
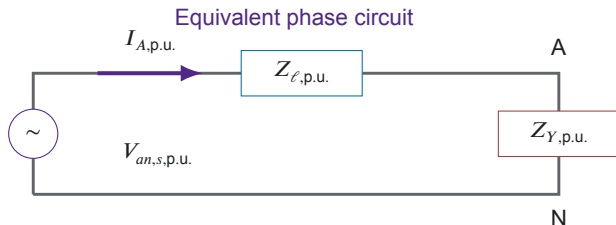
Use $Z_Y = Z_{\Delta}/3$. Keep the same source, line impedances, and terminals A, B, C.

- 2 Normalize the phase circuit.

Use $V_{LN,base} = V_{LL,base}/\sqrt{3}$ and $Z_{base} = V_{LL,base}^2/S_{base,3\phi}$.

Convert $V_{ab,s}$ to $V_{an,s}$ with the -30° angle change before normalizing.

Example 2 workflow: solve and restore



3 Solve the phase circuit.

$$I_{A,p.u.} = \frac{V_{an,s,p.u.}}{Z_{\ell,p.u.} + Z_{Y,p.u.}}$$

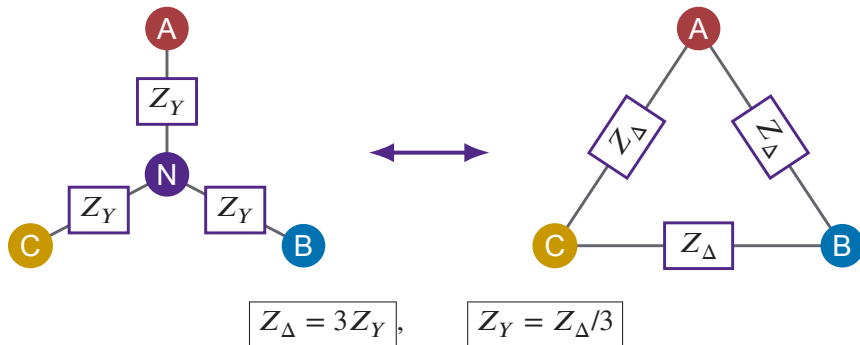
Find $V_{AN,p.u.} = I_{A,p.u.} Z_{Y,p.u.}$. Restore I_A and V_{AN} , then form V_{AB} .

4 Return to the physical Δ .

$$I_{AB} = \frac{V_{AB}}{Z_{\Delta}}$$

Rotate to obtain the other branches. Check $I_A = I_{AB} - I_{CA}$ and total load power.

E2 Step 1: Y–delta conversion



An equivalent transformation preserves terminal voltages, line currents, and total load power.

E2 Step 1: equivalent Y load

$$Z_Y = \frac{Z_\Delta}{3} = \frac{24 + j18}{3} = \boxed{8 + j6 \, \Omega}$$

$$Z_\Delta = 3Z_Y, \quad Z_\ell = 0.08 + j0.32 \, \Omega \text{ per phase}$$

The source and line stay unchanged. Only the load representation changes before normalization.

E2 Step 2: three-phase bases

Voltage and power bases

$$V_{LN,base} = \frac{13.8}{\sqrt{3}} = 7.967434 \text{ kV}$$

$$S_{base,1\phi} = \frac{20}{3} = 6.666667 \text{ MVA}$$

Line-current and impedance bases

$$I_{base} = \frac{20 \times 10^6}{\sqrt{3}(13,800)} = 836.7395 \text{ A}$$

$$Z_{base} = \frac{13.8^2}{20} = \boxed{9.522 \text{ } \Omega}$$

The bases satisfy $Z_{base} = V_{LN,base}/I_{base}$.

E2 Step 2: source phase voltage

Y connection, positive sequence

$$V_{ab,s} = \sqrt{3} V_{an,s} \angle 30^\circ$$

$$I_{\text{line}} = I_{\text{Y branch}}$$

The same voltage relation applies at the load terminals.

Source phase voltage

$$V_{an,s} = \frac{13.8}{\sqrt{3}} \angle (-30^\circ) \text{ kV}$$

$$V_{an,s} = 7.967434 \angle (-30^\circ) \text{ kV}$$

The 30° line-to-phase angle change follows from the abc phasors.

E2 Step 2: normalized data

Quantity	Physical value	Per-unit value
$V_{an,s}$	$7.967434\angle(-30^\circ) \text{ kV}$	$1\angle(-30^\circ)$
Z_ℓ	$0.08 + j0.32 \, \Omega$	$0.008402 + j0.033606$
Z_Y	$8 + j6 \, \Omega$	$0.840160 + j0.630120$

$$V_{an,s,pu} = \frac{V_{an,s}}{V_{LN,base}}, \quad Z_{k,pu} = \frac{Z_k}{Z_{base}}$$

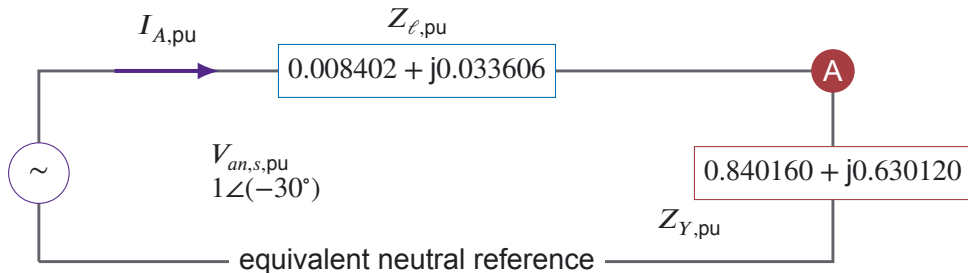
A positive real base changes the magnitude scale and preserves the phasor angle.

E2 Step 2: delta on the same base

$$\frac{Z_{\Delta}}{Z_{\text{base}}} = \frac{24 + j18}{9.522} = 2.520479 + j1.890359$$
$$Z_{Y,\text{p.u.}} = \frac{Z_{\Delta}/3}{Z_{\text{base}}} = \frac{2.520479 + j1.890359}{3} = \boxed{0.840160 + j0.630120}$$

On the same impedance base, normalization does not remove the Δ -Y factor of three.

E2 Step 2: equivalent PU circuit



$$I_{A,pu} = \frac{V_{an,s,pu}}{Z_{\ell,pu} + Z_{Y,pu}}$$

All values on the diagram use the Example 2 bases.

E2 Step 3: line current

$$I_{A,\text{pu}} = \frac{1 \angle (-30^\circ)}{0.848561 + j0.663726} = \boxed{0.347248 - j0.860843}$$

$$I_{A,\text{pu}} = 0.928241 \angle (-68.031743^\circ)$$

$$I_A = I_{A,\text{pu}} I_{\text{base}} = \boxed{776.696 \angle (-68.032^\circ) \text{ A}}$$

Current lags its source phase voltage by 38.032°.

E2 Step 3: equivalent phase voltage

$$V_{AN,pu} = I_{A,pu} Z_{Y,pu} = 0.834178 - j0.504437$$

$$V_{AN,pu} = 0.974839 \angle (-31.161846^\circ)$$

$$V_{AN} = V_{AN,pu} V_{LN,base} = 7.766961 \angle (-31.162^\circ) \text{ kV}$$

The independent route is $V_{AN,pu} = V_{an,s,pu} - I_{A,pu} Z_{\ell,pu}$.

E2 Step 3: load line voltage

$$V_{AB} = \sqrt{3} V_{AN} \angle 30^\circ = \boxed{13.452772 \angle (-1.161846^\circ) \text{ kV}}$$

Phase-voltage normalization

$$V_{AN,\text{pu}} = 0.974839 \angle (-31.162^\circ)$$

Line-voltage normalization

$$V_{AB,\text{pu}} = 0.974839 \angle (-1.162^\circ)$$

Compatible bases give equal per-unit voltage magnitudes. The phasor angles still differ by 30° .

E2 Step 4: delta branch voltages

Each physical Δ branch lies between two load line terminals.

$$V_{AB} = 13.452772 \angle (-1.161846^\circ) \text{ kV},$$

$$V_{BC} = 13.452772 \angle (-121.161846^\circ) \text{ kV},$$

$$V_{CA} = 13.452772 \angle 118.838154^\circ \text{ kV}.$$

Delta branch voltage equals load line voltage. The source line voltage includes the additional line drop.

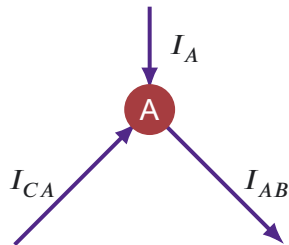
E2 Step 4: delta branch currents

$$I_{AB} = \frac{V_{AB}}{Z_{\Delta}} = \frac{13,452.772 \angle (-1.161846^{\circ})}{30 \angle 36.869898^{\circ}} = \boxed{448.426 \angle (-38.032^{\circ}) \text{ A}}$$

$$I_{BC} = 448.426 \angle (-158.032^{\circ}) \text{ A}, \quad I_{CA} = 448.426 \angle 81.968^{\circ} \text{ A}$$

Branch currents follow the defined directions AB, BC, and CA.

E2 Step 4: line-current check



$$I_A = I_{AB} - I_{CA}$$

$$I_A = 290.556 - j720.301 \text{ A}$$

$$I_A = 776.696 \angle (-68.032^\circ) \text{ A}$$

For this balanced abc delta, $I_A = \sqrt{3} I_{AB} \angle (-30^\circ)$.

E2 Step 4: load power in pu

$$S_{L,\text{p.u.}} = V_{AN,\text{p.u.}} I_{A,\text{p.u.}}^* = |I_{A,\text{p.u.}}|^2 Z_{Y,\text{p.u.}} = 0.723908 + j0.542931$$

$$S_L = S_{L,\text{p.u.}} (20 \text{ MVA}) = \boxed{14.478166 + j10.858624 \text{ MVA}}$$

This PU power uses the total three-phase base. In physical units, the equivalent Y gives $S_L = 3V_{AN}I_A^*$.

E2 Step 4: Y–delta power check

Equivalent Y calculation

$$S_L = 3V_{AN}I_A^* = 3|I_A|^2 Z_Y$$

$$|I_A| = 776.696 \text{ A}$$

$$S_L = 14.478166 + j10.858624 \text{ MVA}$$

Physical delta calculation

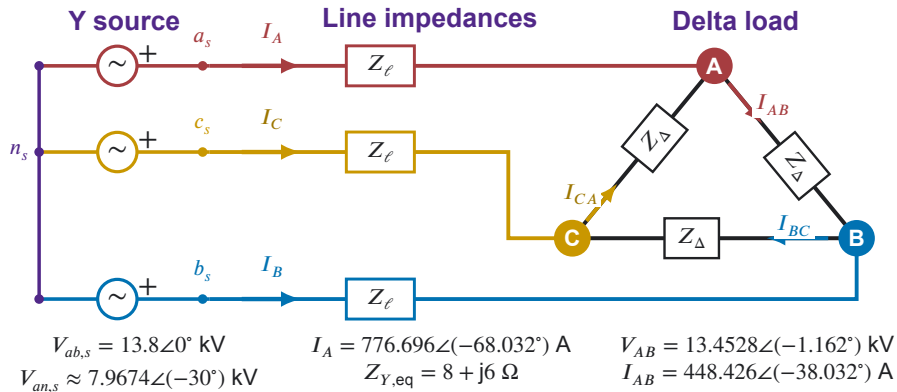
$$S_L = 3V_{AB}I_{AB}^* = 3|I_{AB}|^2 Z_{\Delta}$$

$$|I_{AB}| = 448.426 \text{ A}$$

Each power product pairs a branch voltage with the current through that same branch.

Example 2: circuit answer

Original physical Δ circuit; $Z_{\Delta} = 24 + j18 \Omega$ per branch. Equivalent Y: $Z_Y = 8 + j6 \Omega$.



$S_L = 14.478166 + j10.858624 \text{ MVA}$. Other balanced phases follow by $\pm 120^\circ$ rotation.

$I_A = I_{AB} - I_{CA}$; line current and total power agree with the normalized Y model.

Two examples: base conventions

Example 1: single phase

$$I_{\text{base}} = S_{\text{base}}/V_{\text{base}}$$

$$Z_{\text{base}} = V_{\text{base}}^2/S_{\text{base}}$$

Transformer ratios set the voltage bases. Actual loop currents use their local bases.

Example 2: three phases

$$I_{\text{base}} = \frac{S_{\text{base},3\phi}}{\sqrt{3}V_{LL,\text{base}}}$$

$$Z_{\text{base}} = \frac{V_{LL,\text{base}}^2}{S_{\text{base},3\phi}}$$

$Z_Y = Z_{\Delta}/3$ sets the phase model.
Physical Δ branches use line voltages.

Connection conversion defines the circuit; normalization defines the scales used to solve it.